

Differential Calculus

*Home work sheet # 4

Maxima and minima

1. Find (a) the open intervals on which f is increasing, (b) the open intervals on which f is decreasing, (c) the open intervals on which f is concave up, (d) the open intervals on which f is concave down and (e) the x - coordinate of all inflection points.

$$(i) f(x) = x^2 - 5x + 6 \quad (ii) f(x) = 5 + 12x - x^3$$

$$(iii) f(x) = x^4 - 8x^2 + 16 \quad (iv) f(x) = \frac{x^2}{x^2 + 2} \quad (v) f(x) = \sqrt[3]{x + 2}.$$

2. Locate the critical numbers and identify which critical numbers correspond to stationary points.

$$(i) f(x) = x^3 + 3x^2 - 9x + 1 \quad (ii) f(x) = x^4 - 6x^2 - 3$$

$$(iii) f(x) = \frac{x}{x^2 + 2} \quad (iv) f(x) = x^{2/3}$$

$$(v) f(x) = x^{1/3}(x + 4) \quad (vi) f(x) = \cos 3x.$$

3. Find the relative extrema (maxima/ minima) using both the first and second derivative tests.

$$(i) f(x) = 2x^3 - 9x^2 + 12x \quad (ii) f(x) = \frac{x}{2} - \sin x, \quad 0 < x < 2\pi.$$

4. Use the given derivative to find all critical numbers of $f(x)$ and determine whether a relative maximum, relative minimum, or neither occurs there.

$$(i) f'(x) = x^3(x^2 - 5) \quad (ii) f'(x) = \frac{x^2 - 1}{x^2 + 1}.$$

**These problems are for the students only as home work. Search the reference books for more examples.*